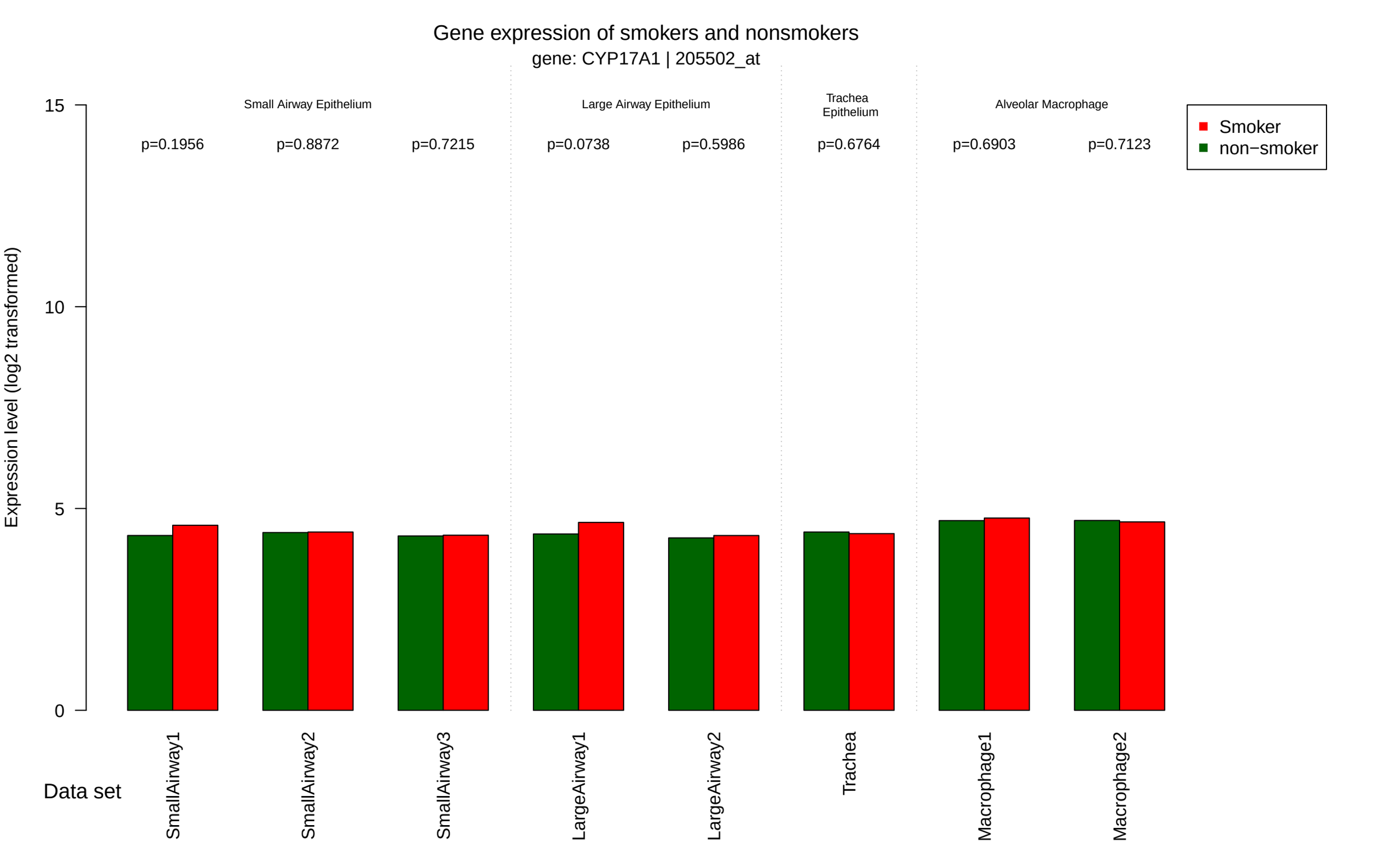
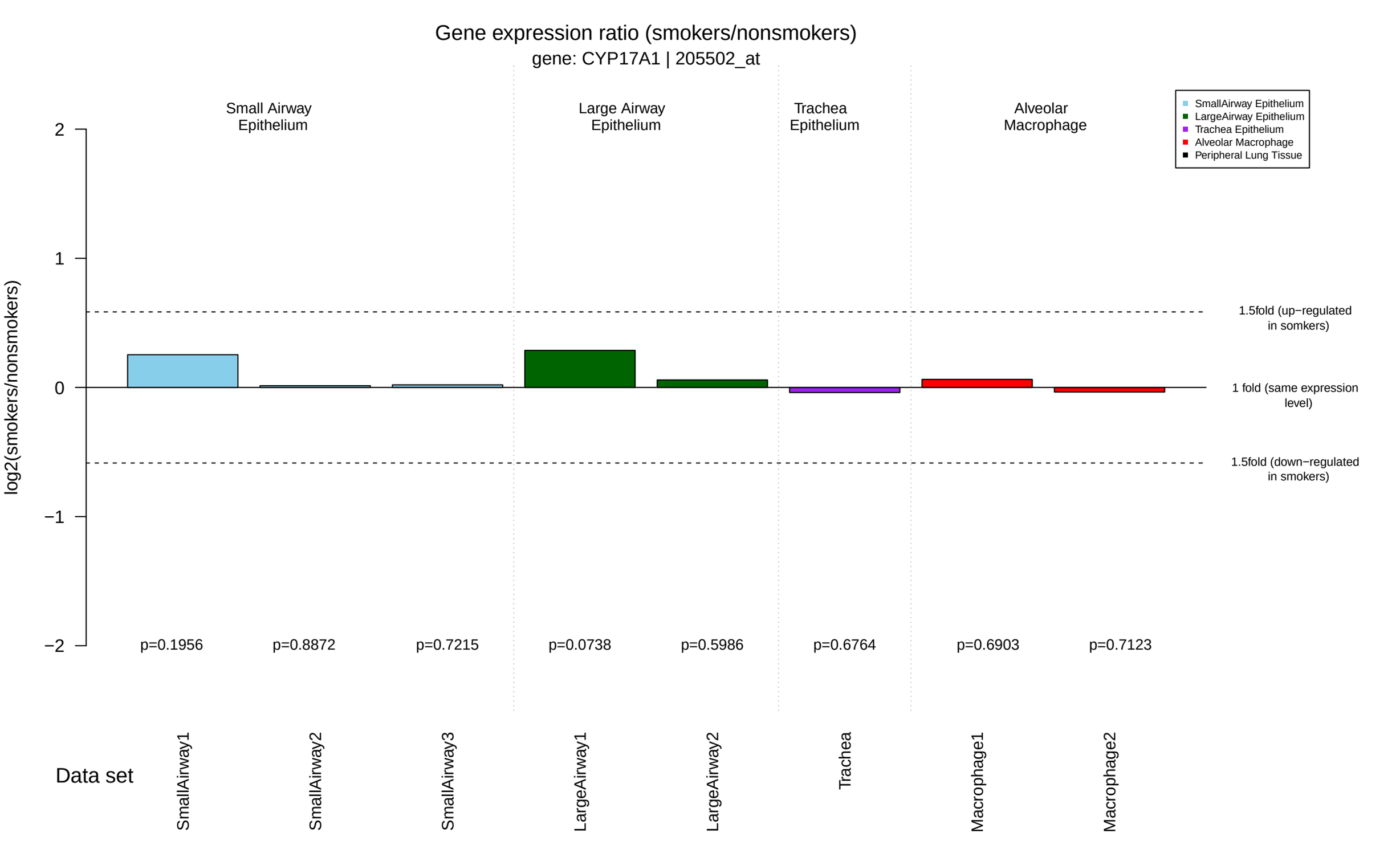
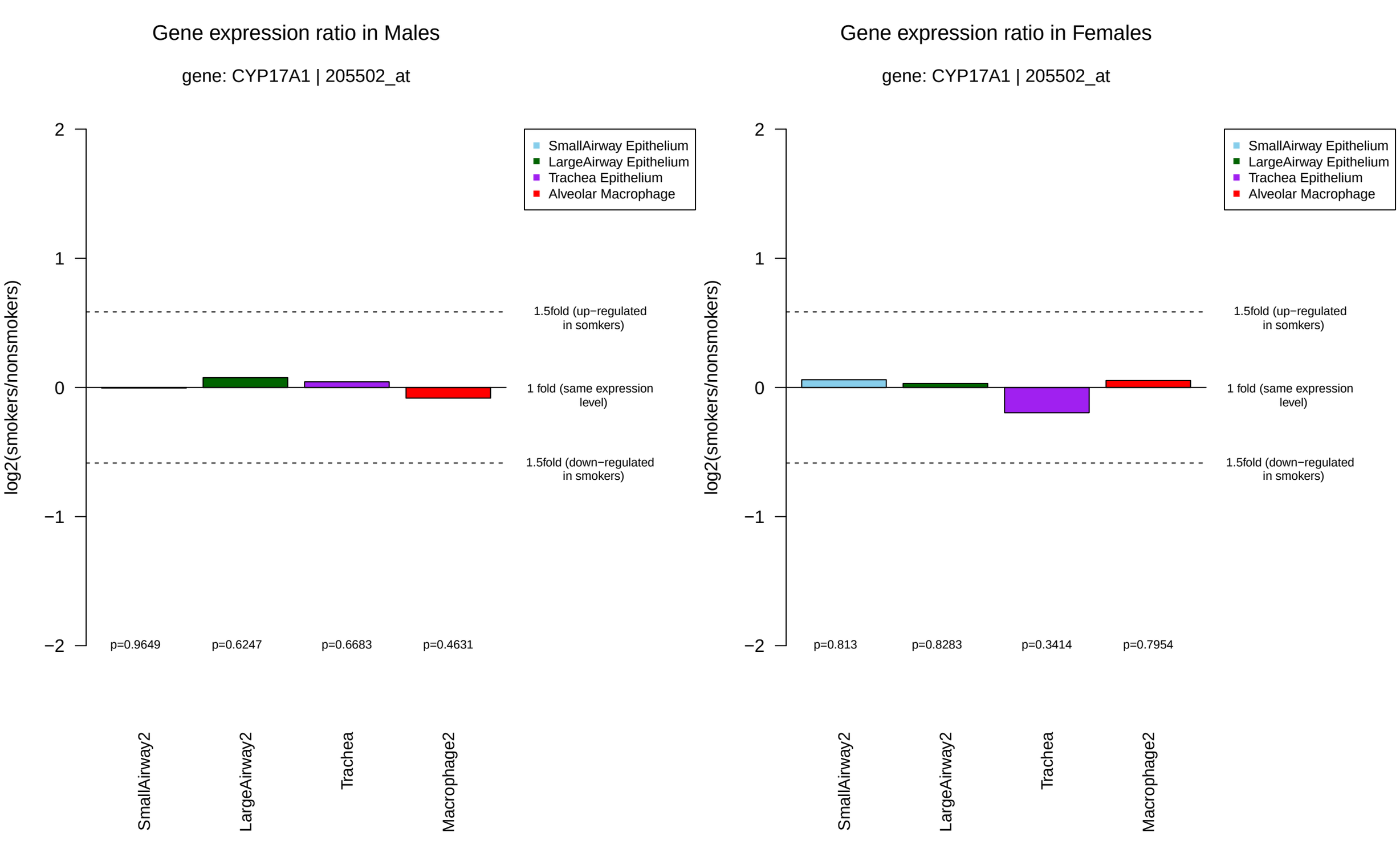
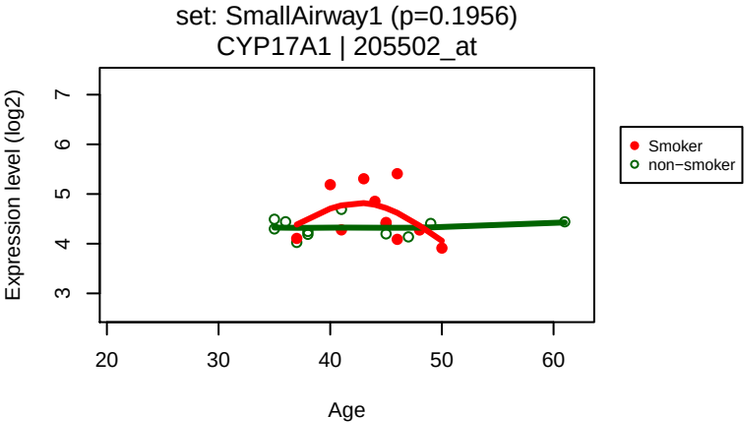
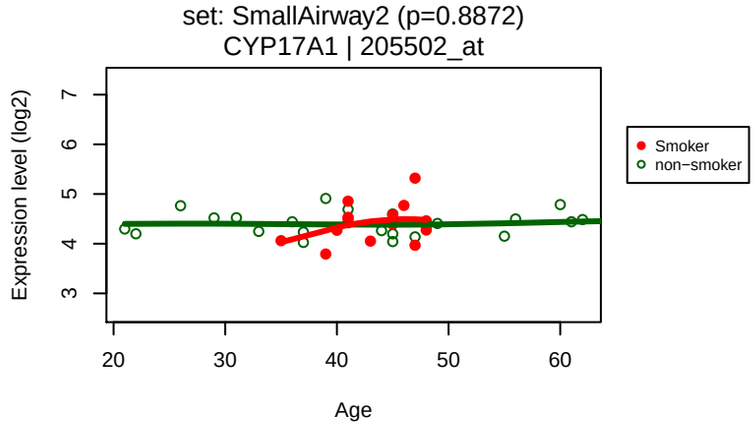
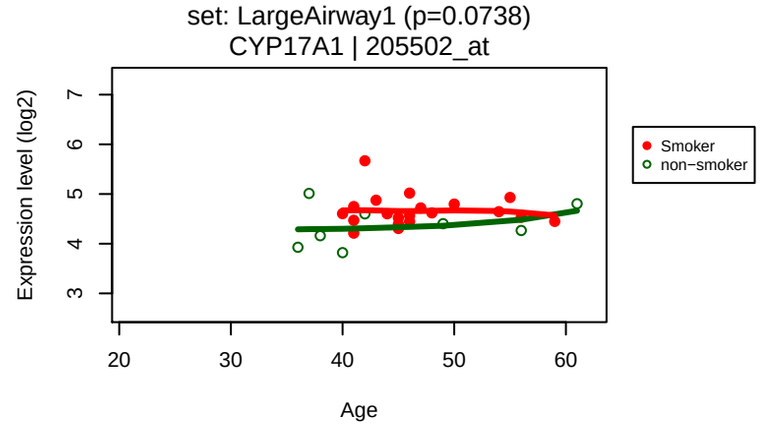

Gene expression of smokers and nonsmokers
gene: CYP17A1 | 205502_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
p=0.1956
p=0.8872
p=0.7215
p=0.0738
p=0.5986
p=0.6764
p=0.6903
p=0.7123
Smoker
non−smoker
Expression level (log2 transformed)
15
10
5
0
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

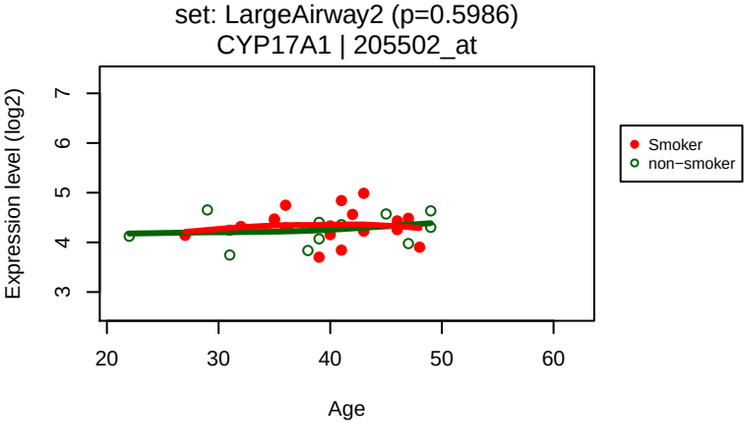

Gene expression ratio (smokers/nonsmokers)
gene: CYP17A1 | 205502_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Peripheral Lung Tissue
log2(smokers/nonsmokers)
2
1
0
−1
−2
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.1956
p=0.8872
p=0.7215
p=0.0738
p=0.5986
p=0.6764
p=0.6903
p=0.7123
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

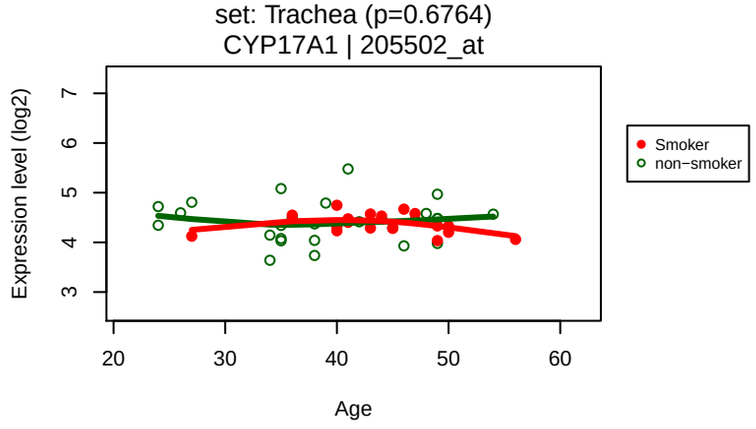

Gene expression ratio in Males
gene: CYP17A1 | 205502_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.9649
p=0.6247
p=0.6683
p=0.4631
SmallAirway2
LargeAirway2
Trachea
Macrophage2
Gene expression ratio in Females
gene: CYP17A1 | 205502_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.813
p=0.8283
p=0.3414
p=0.7954
SmallAirway2
LargeAirway2
Trachea
Macrophage2

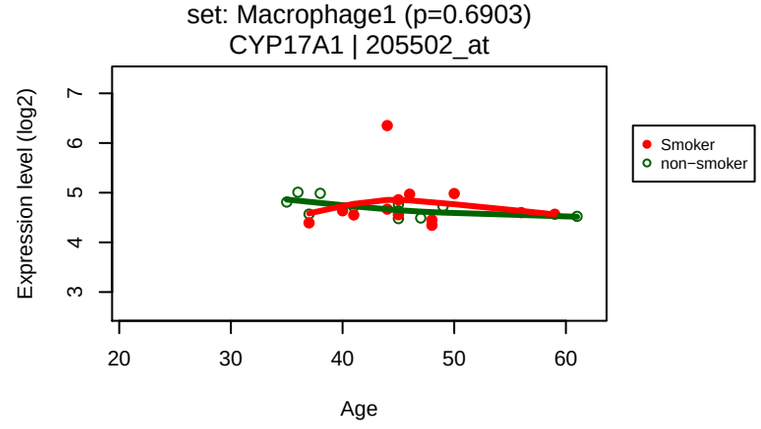

set: SmallAirway1 (p=0.1956)
CYP17A1 | 205502_at
Expression level (log2)
Age
Smoker
non−smoker

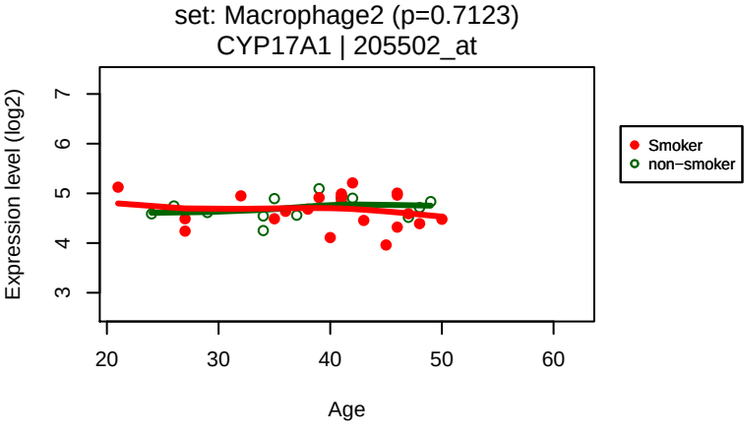

set: SmallAirway2 (p=0.8872)
CYP17A1 | 205502_at
Expression level (log2)
Age
Smoker
non−smoker

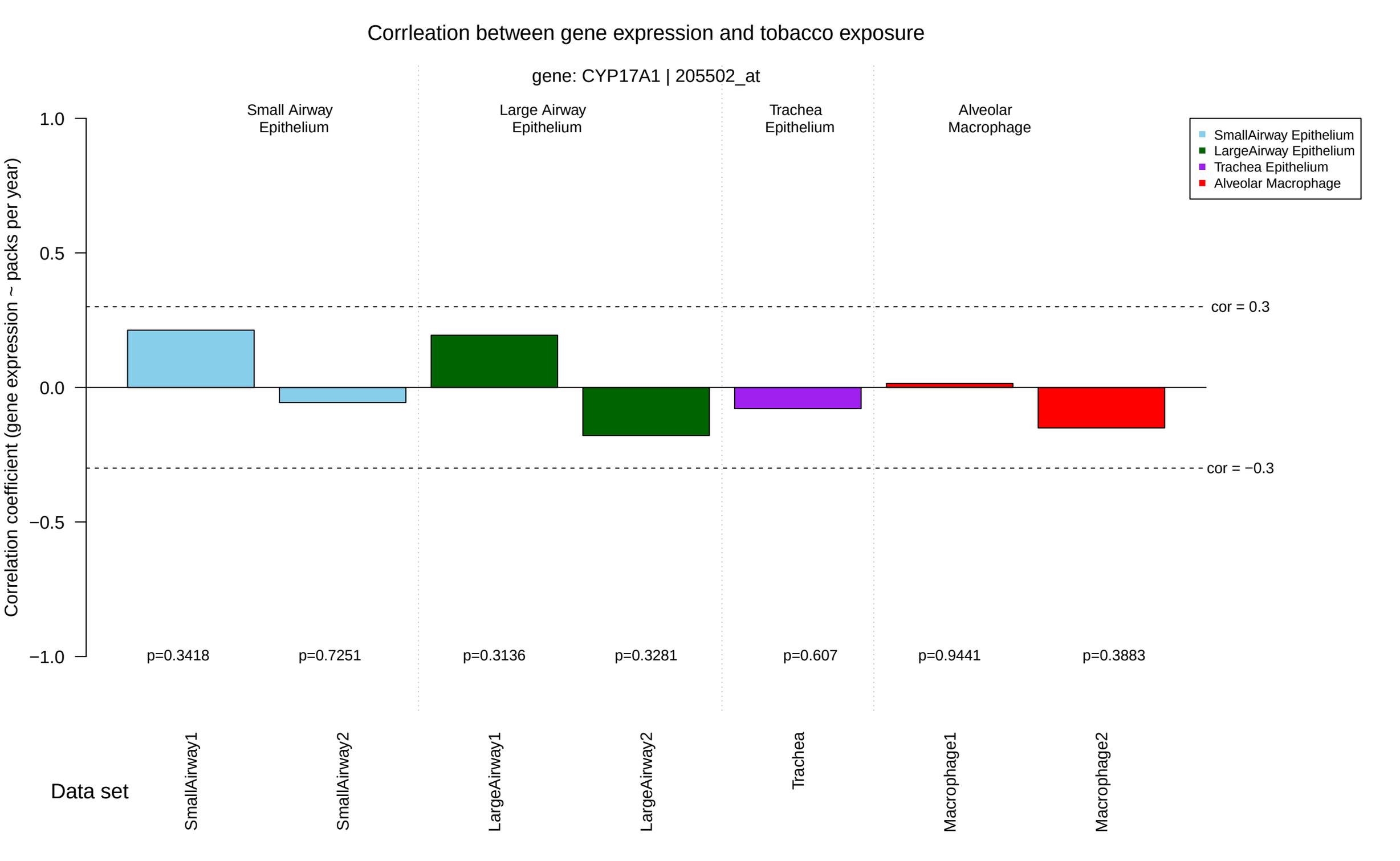

set: LargeAirway1 (p=0.0738)
CYP17A1 | 205502_at
Expression level (log2)
Age
Smoker
non−smoker


set: LargeAirway2 (p=0.5986)
CYP17A1 | 205502_at
Expression level (log2)
Age
Smoker
non−smoker


set: Trachea (p=0.6764)
CYP17A1 | 205502_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage1 (p=0.6903)
CYP17A1 | 205502_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage2 (p=0.7123)
CYP17A1 | 205502_at
Expression level (log2)
Age
Smoker
non−smoker


Corrleation between gene expression and tobacco exposure
gene: CYP17A1 | 205502_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Correlation coefficient (gene expression ~ packs per year)
1.0
0.5
0.0
−0.5
−1.0
cor = 0.3
cor = −0.3
p=0.3418
p=0.7251
p=0.3136
p=0.3281
p=0.607
p=0.9441
p=0.3883
Data set
SmallAirway1
SmallAirway2
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2